

1MT495 Current Issues in Applied International Economics: Neoclassical Approach in International Trade

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Outline

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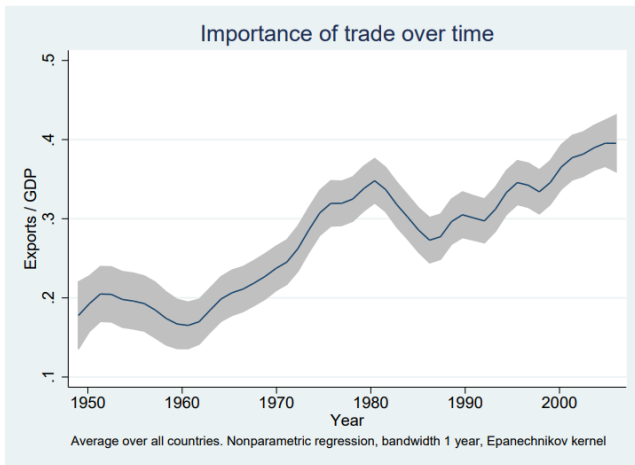
General Information about my course part

- My goal is to give you a broad overview of different models and approaches.
- There is a lot of mathematics in economics but you don't need to dive into every detail; **this is not a math course. You do need to understand the math, but math can help you understand the intuition.**
- I will focus on the **main ideas and ways of thinking in modern international economics.**
- Most of the material is not my own. I have collected and adapted lecture notes from people much smarter than me.
- **Deeper Dive:** Always check the original sources for more technical details or deeper exploration.

Introduction to Neoclassical Trade Theory

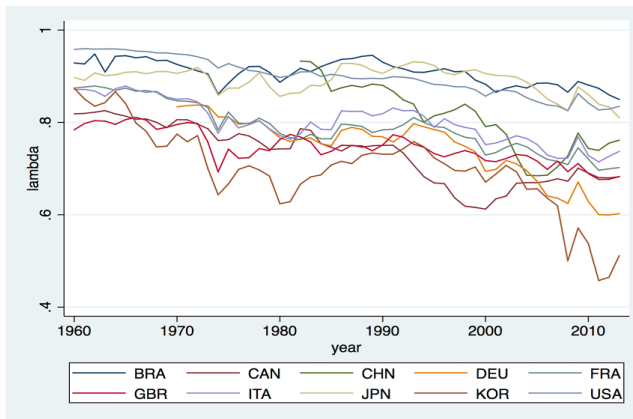
- We begin by revisiting the **foundations** of trade theory, building upon classical insights.
- Neoclassical theory emphasizes **efficiency**, **gains from trade**, and resource allocation.
- **Key Models:**
 - **Ricardian Model:** Trade driven by differences in **relative productivity** (**comparative advantage**).
 - **Heckscher-Ohlin (HO) Model:** Trade driven by differences in **factor endowments** (e.g., capital, labor).
- **Standard Assumptions:** **Perfect competition**, **free trade**, and **balanced trade** in equilibrium.

Empirical evidence: Importance of Trade Over Time



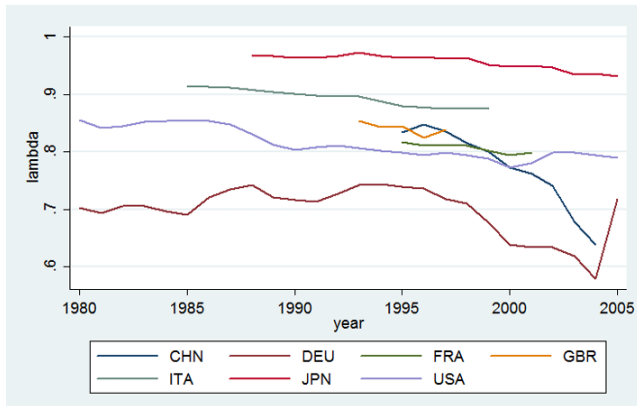
Source: Treb Allen lecture notes

Empirical evidence: Spending on Domestic Goods



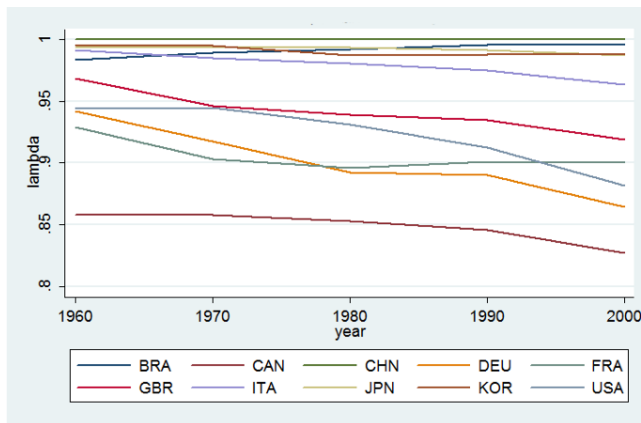
Source: Costas Arkolakis - A Primer on Gains From Openness

Empirical evidence: Domestic Production



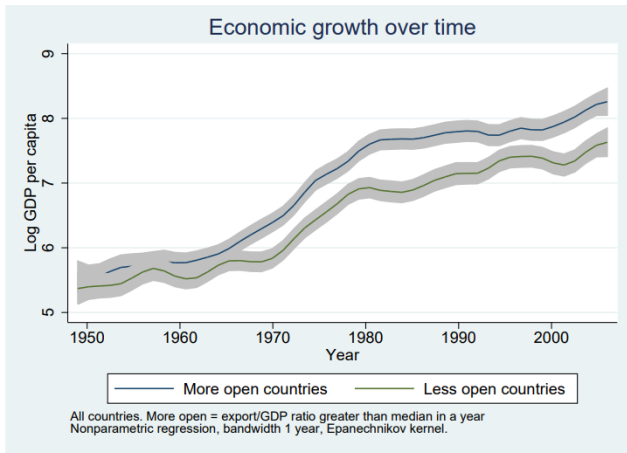
Source: Costas Arkolakis - A Primer on Gains From Openness

Empirical evidence: Native Born Share of US Population



Source: Costas Arkolakis - A Primer on Gains From Openness

Empirical evidence: GDP per Capita Over Time



Source: Treb Allen lecture notes

Ricardian Model: Main Ideas

- **Trade Driver:** Differences in **relative labor productivity** (technology) across countries.
- **Core Concept: Comparative Advantage (CA).** A country should export the good for which it has a lower **opportunity cost** of production.
- **Specialization:** Trade leads to specialization based on CA, not **Absolute Advantage**. This means a country benefits from specialization even if it is less efficient in all goods.
- **Gains from Trade:** The increase in world output and consumption resulting from this specialization.

Ricardian Model: Assumptions

- **Structure:** Two countries, two goods, **labor (L) as the sole factor of production.**
- **Technology:** Fixed **unit labor requirements** (α_i) (the inverse of productivity). **Constant Returns to Scale (CRS).**
- **Markets:** **Perfect competition.**
- **Mobility:** Labor is mobile domestically but **immobile internationally**. All laborers are identical.

Ricardian Model: Autarky Derivation (PPF)

- **Full Employment Constraint:** The total labor endowment (L) is fully utilized:

$$\alpha_{wine} Q_{wine} + \alpha_{cloth} Q_{cloth} \leq L$$

- **Production Possibility Frontier (PPF):** Solving the constraint for Q_{wine} :

$$Q_{wine} = \left(\frac{L}{\alpha_{wine}} \right) - \left(\frac{\alpha_{cloth}}{\alpha_{wine}} \right) Q_{cloth}$$

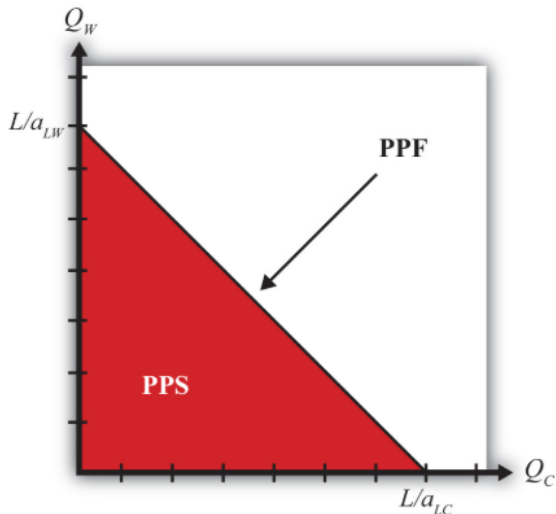
- **Opportunity Cost and Slope:**

- The slope of the PPF is: $\text{Slope} = -\frac{\alpha_{cloth}}{\alpha_{wine}}$.
- This slope represents the **opportunity cost** of one unit of cloth in terms of wine (i.e., how many units of wine must be given up to produce one more unit of cloth).

- **Autarky Equilibrium:** Production and consumption occur where the PPF is tangent to the country's indifference curve.

- **Autarky Relative Price:** The relative price equals the opportunity cost: $p_{cloth}/p_{wine} = \alpha_{cloth}/\alpha_{wine}$.

Ricardian Model: PPF



Note: The linear PPF is a direct consequence of **one factor** and **CRS**.

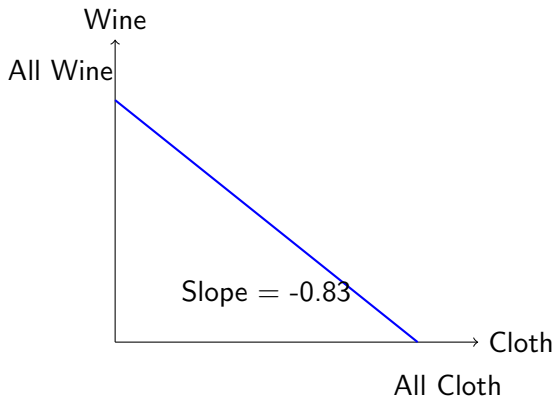
Ricardian Model: Wages and Prices

- **Competitive Pricing:** In a competitive market, price equals cost. With labor as the sole factor: $p_i = w \cdot \alpha_i$, where w is the wage.
- **Normalization:** If we normalize the home wage ($w = 1$), then $p_i = \alpha_i$.
- **Wage Determination:** In the free-trade equilibrium, workers produce the good where the real wage ($\frac{p_i}{\alpha_i}$) is maximized.
- **Implication:** Trade equalizes effective prices ($\frac{p_i}{\alpha_i}$) but **absolute wages** (or relative wages between countries) are determined by **absolute productivity** ratios.
- **Specialization: Complete specialization** is a strong implication due to the linear technology and lack of factor substitutability.

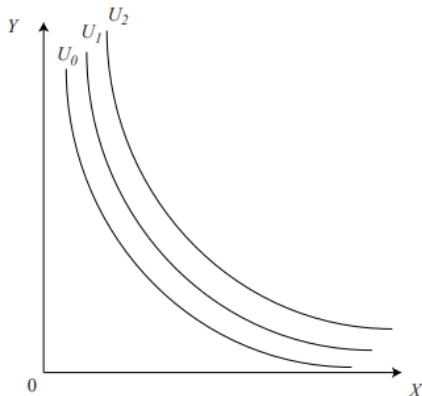
Ricardian Model: England-Portugal Example

- **Unit Labor Requirements (α_i):**
 - Portugal (P): $\alpha_{cloth} = 90$, $\alpha_{wine} = 80$
 - England (E): $\alpha_{cloth} = 100$, $\alpha_{wine} = 120$
- **Absolute Advantage:** Portugal has an absolute advantage in both goods (lower α_i).
- **Autarky Relative Prices (p_{cloth}/p_{wine}):**
 - Portugal: $90/80 = \mathbf{1.125}$ (Cloth is relatively expensive).
 - England: $100/120 \approx \mathbf{0.833}$ (Cloth is relatively cheap).
- **Comparative Advantage (CA):**
 - Portugal has CA in **wine** (its $\alpha_{cloth}/\alpha_{wine}$ is higher than England's, meaning its cost of wine is lower in terms of cloth).
 - England has CA in **cloth** (its $\alpha_{cloth}/\alpha_{wine}$ is lower than Portugal's, meaning its cost of cloth is lower in terms of wine).

Ricardian Model: England-Portugal Example (Portugal)



Ricardo's Model: Demand side



On an indifference curve, $MRS = -\frac{dY}{dX} = \frac{\partial U/\partial X}{\partial U/\partial Y}$

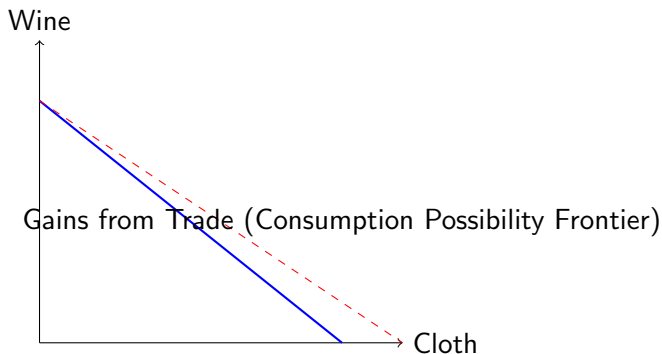
At the consumer's optimum : $\frac{P_X}{P_Y} = MRS$

Ricardian Model: Autarky and Trade Equilibrium

- **Trade Price:** The world relative price (p_{cloth}/p_{wine}) will settle **between** the two autarky prices: 0.833 $(\alpha_{cloth})/\alpha_{wine} < p_{cloth}/p_{wine} < 1.125 (\alpha_{cloth}^*)/\alpha_{wine}^*)$.
- **Specialization:** Each country specializes in its CA good.
 - **England** specializes in Cloth.
 - **Portugal** specializes in Wine.
- **Gains from Trade:** The country's **Consumption Possibility Frontier (CPF)** expands beyond its PPF, allowing higher welfare.

Ricardian Model: Trade

Trade separates Production Possibilities from Consumption Possibilities.



- **Production** stays on the **PPF**.
- **Consumption** is now on the **CPF**, indicating a higher real income.

Ricardian Model: Implications and Extensions

- **Key Implications:**

- Trade increases **aggregate output** and welfare.
- Trade is **not a zero-sum game** (absolute gains for all participants).
- The model implies **balanced trade** in equilibrium.

- **Criticisms:** Too simple; ignores multiple factors, income distribution, and intra-industry trade.

- **Modern Extensions:**

- **Eaton-Kortum (2002):** A probabilistic Ricardian model that incorporates **geography** (distance, trade costs) and a continuum of goods, offering a much better empirical fit (e.g., with **Gravity Equations**).
- The EK model shows that countries trade based on a mix of productivity, costs, and distance not just with the absolute most efficient producer.

Heckscher-Ohlin Model: Historical Context factory

- **Origin:** Developed by Eli Heckscher and Bertil Ohlin in the early 20th century.
- **Key Difference from Ricardo:** The HO model moves beyond a single factor (labor) to incorporate **multiple factors** (typically labor L and capital K).
- **Trade Driver:** Differences in **relative factor endowments** (e.g., country A has more K relative to L than country B).

Heckscher-Ohlin Model: Assumptions

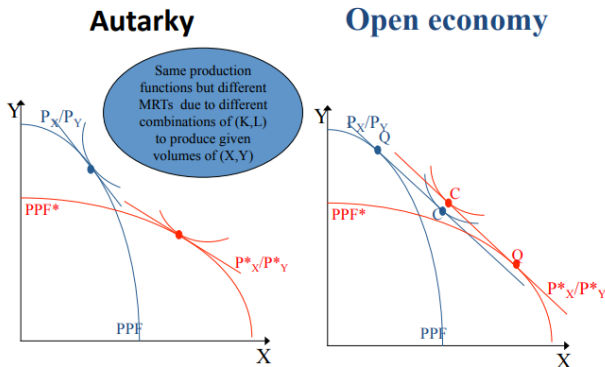
- **Structure:** Two countries (H, F), two goods (X, Y), two factors (L, K).
- **Technology:** **Identical technologies** and **Constant Returns to Scale** (CRS) across countries.
- **Factors:** Mobile **within** countries, **immobile across** countries.
- **Markets/Preferences:** Perfect competition, identical **homothetic preferences**.
- **Critical Technical Assumption:** No **Factor Intensity Reversals** (FIRs) - one good is always labor-intensive relative to the other, regardless of factor prices.

Model Setup

Consider two goods:

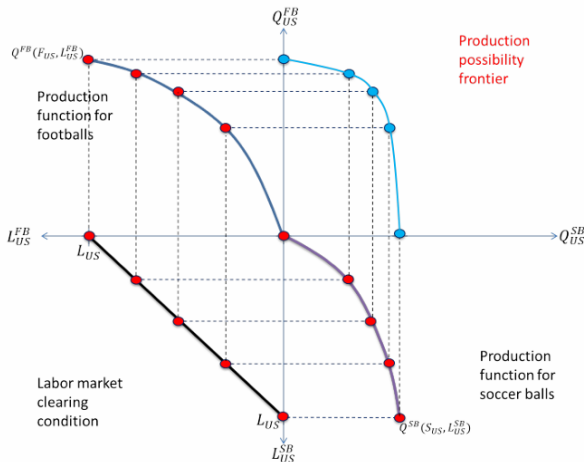
- Good X: **Labor-intensive** (uses a high L/K ratio in production).
- Good Y: **Capital-intensive** (uses a high K/L ratio in production).
- **Production Functions:** $Q_j = Q_j(L_j, K_j)$ for $j = X, Y$.
- **Endowments:** Total factor supply $\bar{L}$ and $\bar{K}$ are fixed in each country.
- **Full Employment Constraint:** $\bar{L} = L_X + L_Y$ and $\bar{K} = K_X + K_Y$.

Heckscher-Ohlin Model: PPF



Note: Unlike the Ricardian PPF, the HO PPF is **bowed out (concave)** due to **diminishing returns** (moving factors across sectors) and the possibility of **factor substitution**.

Heckscher-Ohlin Model: PPF Derivation



Source: Treb Allen notes

Heckscher-Ohlin Model: Key Theorems

- **Heckscher-Ohlin (HO)**: A country exports the good that intensively uses its **relatively abundant factor**. (Trade Driver)
- **Factor Price Equalization (FPE)**: Free trade in goods can **equalize factor prices** (wages w and rent r) across countries. (Trade Result)
- **Stolper-Samuelson (SS)**: An increase in the relative price of a good increases the real return to the factor used **intensively** in its production, and lowers the return to the other factor. (Distributional Result)
- **Rybczynski (Ryb)**: At constant goods prices, an increase in a factor endowment leads to a disproportionate expansion of the output of the good that uses that factor **intensively**, and an absolute contraction of the other good's output. (Production Result)

Production and Unit Input Functions

- **Production Functions:** Assume Cobb-Douglas (for tractability):

$$Q_j = L_j^{\theta_{Lj}} K_j^{\theta_{Kj}}, \quad \text{where } \theta_{Lj} + \theta_{Kj} = 1$$

- **Cost Minimization:** For good j , firms minimize $\text{Cost} = wL_j + rK_j$ subject to $Q_j(L_j, K_j) = 1$.
- **Optimal Factor Ratio:** For good X (labor-intensive, $\theta_{LX} = \beta > 0.5$):

$$\frac{K_X}{L_X} = \frac{w}{r} \cdot \frac{1 - \beta}{\beta}$$

- **Unit Inputs (a_{ij}):** The factor input required per unit of output, derived by substituting the optimal ratio back into the production function:

$$a_{LX} = \beta^{1-\beta} (1 - \beta)^{\beta-1} \left(\frac{r}{w} \right)^{1-\beta}$$

$$a_{KX} = \beta^{-\beta} (1 - \beta)^{\beta} \left(\frac{r}{w} \right)^{-\beta}$$

Note: The a_{ij} coefficients are a function of the factor price ratio w/r .

Equilibrium and Duality: The Matrix Structure

The HO model relies on a system of two equations governing factor use and two equations governing costs.

- **Zero-Profit (Competitive Pricing):** Price equals unit cost.

$$p_X = w a_{LX} + r a_{KX}, \quad p_Y = w a_{LY} + r a_{KY}$$

In vector form: $\mathbf{p} = \mathbf{A}^\top \mathbf{w}$, where $\mathbf{w} = (w, r)^\top$.

- **Full Employment:** Endowments (L, K) constrain output (Q_X, Q_Y) .

$$L = a_{LX} Q_X + a_{LY} Q_Y, \quad K = a_{KX} Q_X + a_{KY} Q_Y$$

In vector form: $\mathbf{V} = \mathbf{A} \mathbf{Q}$, where $\mathbf{V} = (L, K)^\top$.

Input Coefficient Matrix ($\mathbf{A}$):

$$\mathbf{A} = \begin{pmatrix} a_{LX} & a_{LY} \\ a_{KX} & a_{KY} \end{pmatrix}$$

Rows are factors (L, K) , columns are goods (X, Y) . The element a_{ij} is the amount of factor i required to produce one unit of good j .

Intensity Condition: X is labor-intensive, implying

$$\det \mathbf{A} = a_{LX} a_{KY} - a_{LY} a_{KX} > 0.$$

Factor Price Equalization (FPE) Derivation

The FPE theorem is a direct result of the zero-profit condition.

- **Derivation:** Inverting the competitive pricing system $\mathbf{p} = \mathbf{A}^\top \mathbf{w}$ gives factor prices as a function of goods prices and technology:

$$\begin{pmatrix} w \\ r \end{pmatrix} = (\mathbf{A}^\top)^{-1} \begin{pmatrix} p_X \\ p_Y \end{pmatrix}$$

- **Conclusion:** If countries have:
 - 1 **Identical Technology** (same $\mathbf{A}$ matrix).
 - 2 **Free Trade** (same goods prices $\mathbf{p}$).
 - 3 **Diversification** (produce both goods).

Then, the factor prices (w, r) are uniquely determined by the global goods prices and must be **equalized** across countries.

Stolper-Samuelson (SS) Theorem: Foundations

The Zero-Profit Condition and Cost Shares

- **Core Idea:** SS demonstrates the **Magnification Effect** of a goods price change on real factor returns.
- **Implication:** Trade affects **income distribution**. Opening to trade benefits the owners of the country's relatively abundant factor.
- **Cost Share Matrix (Θ):** This matrix defines the weight of each factor's cost in a good's final price:

$$\Theta = \begin{pmatrix} \theta_{XL} & \theta_{XK} \\ \theta_{YL} & \theta_{YK} \end{pmatrix}$$

where the cost share of factor i in good j is:

$$\theta_{ij} = \frac{\text{Cost of factor } i \text{ per unit of } j}{p_j} \quad \text{e.g., } \theta_{XL} = \frac{w a_{LX}}{p_X}$$

- **Price Constraint:** Under perfect competition, price equals the sum of factor costs:

$$p_j = w a_{Lj} + r a_{Kj} \quad \implies \quad \theta_{jL} + \theta_{jK} = 1$$

Stolper-Samuelson (SS) Theorem: Log-Differential Derivation

Relating Price Changes ($\hat{p}$) to Factor Return Changes ($\hat{w}, \hat{r}$)

- **Starting Point:** Differentiate the zero-profit condition for good X:

$$dp_X = d(wa_{LX}) + d(ra_{KX}) = a_{LX}dw + wda_{LX} + a_{KX}dr + rda_{KX}$$

- **Envelope Theorem:** Under cost minimization, the change in unit inputs (da_{ij}) in response to factor price changes (dw, dr) is zero. This greatly simplifies the expression:

$$dp_X = a_{LX}dw + a_{KX}dr$$

- **Log-Differential Form:** Dividing by p_X and using cost shares (θ_{ij}):

$$\hat{p}_X = \frac{a_{LX}dw}{p_X} + \frac{a_{KX}dr}{p_X} = \theta_{XL}\hat{w} + \theta_{XK}\hat{r}$$

- **Vector Equation:** This relationship holds for both goods:

$$\begin{pmatrix} \hat{p}_X \\ \hat{p}_Y \end{pmatrix} = \Theta \begin{pmatrix} \hat{w} \\ \hat{r} \end{pmatrix}$$

Stolper-Samuelson (SS) Theorem: Magnification Result

Inverting the Matrix and the Outcome

- **Inverse Relationship:** Factor price changes are determined by the inverse of the cost share matrix multiplied by the goods price changes:

$$\begin{pmatrix} \hat{w} \\ \hat{r} \end{pmatrix} = \Theta^{-1} \begin{pmatrix} \hat{p}_X \\ \hat{p}_Y \end{pmatrix}$$

The inverse matrix is:

$$\Theta^{-1} = \frac{1}{\det \Theta} \begin{pmatrix} \theta_{YK} & -\theta_{XK} \\ -\theta_{YL} & \theta_{XL} \end{pmatrix}$$

- **Magnification Example:** If p_X increases ($\hat{p}_X > 0$) and p_Y is constant ($\hat{p}_Y = 0$), and X is labor-intensive ($\det \Theta > 0$):
 - $\hat{w} = \frac{\theta_{YK}}{\det \Theta} \hat{p}_X$. Since $\theta_{YK} > \det \Theta$, we have $\hat{w} > \hat{p}_X > 0$.
 - $\hat{r} = -\frac{\theta_{YL}}{\det \Theta} \hat{p}_X < 0$.
- **Real Return Conclusion:** The real return to the factor used **intensively** in the favored industry (**Labor**) rises in terms of both goods ($\hat{w} - \hat{p}_X > 0$ and $\hat{w} - \hat{p}_Y > 0$), while the real return to the other factor (**Capital**) falls in terms of both goods.

Rybczynski Theorem (Derivation)

The Rybczynski theorem is a consequence of the full employment system.

- **Derivation:** Inverting the full employment system $\mathbf{V} = \mathbf{A}\mathbf{Q}$:

$$\begin{pmatrix} Q_X \\ Q_Y \end{pmatrix} = \mathbf{A}^{-1} \begin{pmatrix} L \\ K \end{pmatrix}$$

At fixed prices, $\mathbf{A}$ is constant. Differentiating shows the effect of endowment changes ($d\mathbf{V}$) on output changes ($d\mathbf{Q}$):

$$d \begin{pmatrix} Q_X \\ Q_Y \end{pmatrix} = \mathbf{A}^{-1} d \begin{pmatrix} L \\ K \end{pmatrix}$$

where $\mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}} \begin{pmatrix} a_{KY} & -a_{LY} \\ -a_{KX} & a_{LX} \end{pmatrix}$.

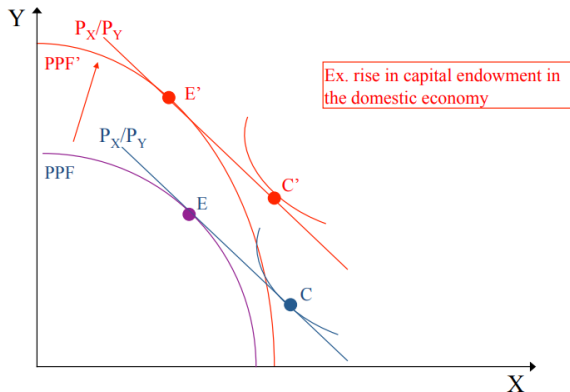
- **Magnification Effect:** For an increase in Labor ($dL > 0, dK = 0$):

$$dQ_X = \frac{a_{KY}}{\det \mathbf{A}} dL > 0 \quad \text{and} \quad dQ_Y = -\frac{a_{LY}}{\det \mathbf{A}} dL < 0$$

The output of the labor-intensive good (X) expands, and the output of the capital-intensive good (Y) contracts absolutely.

Rybczynski Theorem

Shifts the PPF outward, with a strong bias toward the intensive good.



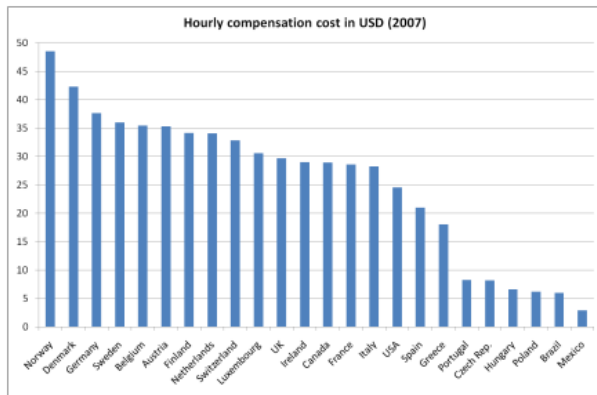
Summary Table of Key HO Theorems

Theorem	Driving Force	Key Result
HO	Factor Endowment Difference	Trade pattern (Export abundant-intensive good)
FPE	Free Trade in Goods	Factor price equalization (w, r equalized)
Stolper-Samuelson	Goods Price Change	Factor returns (Intensive factor gains in real terms)
Rybczynski	Factor Endowment Change	Output pattern (Intensive good output expands)

Heckscher-Ohlin Model: Empirical Evidence and Criticisms

- **Successes:** Predicts trade patterns between countries with **dissimilar factor endowments** (e.g., North-South trade). Explains specialization, though **not complete specialization**.
- **Leontief Paradox (1953):** Found the capital-abundant US was exporting labor-intensive goods a major early challenge to the model.
- **Criticisms:**
 - **Assumptions:** Identical technologies and FPE fail empirically.
 - **Trade Patterns:** Ignores **intra-industry trade** (e.g., trade in cars between similar countries).
 - **Factors:** Poor performance with simple L and K ; better when L is disaggregated (e.g., skilled vs. unskilled).
 - Resource-rich countries export commodities intensive in those resources.
 - Fails to explain persistent trade imbalances.

Empirical evidence: No Wage Equalization

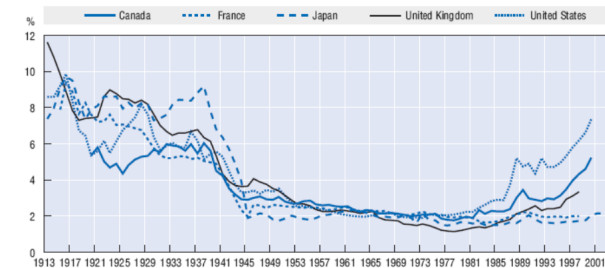


Source : US Bureau of Labor Statistics (August 2009)

Source: Isabelle Mjean lecture notes

Empirical evidence: Inequality

Top 0.1% income share in 5 OECD countries, 1913-2001



Source: Piketty and Saez (2006).

Source: Isabelle Mjean lecture notes

Relation Between Neoclassical Models

- **Ricardo**: Trade is driven by **differences in relative productivity (technology)**. Implies **complete specialization**.
- **Heckscher-Ohlin**: Trade is driven by **differences in factor endowments**. Allows for **incomplete specialization**.
- **Synthesis**: HO can be seen as a generalization. If HO only had one factor (L), it would simplify to the Ricardian model.
- **Gains/Distribution**: Both models show **aggregate gains from trade**.
 - **Ricardo**: All labor is identical, so all workers benefit equally (no income distribution effect).
 - **HO**: Trade causes a change in factor prices (w, r), leading to **winners and losers** within a country (via Stolper-Samuelson).

Conclusions

- **Neoclassical Core:** These models establish that trade yields **aggregate gains** through specialization (comparative advantage and factor endowments).
- **Specialization:** Leads to higher global efficiency and allows countries to consume beyond their PPF.
- **Empirical Challenges:** Unrealistic assumptions (FPE, no tech gaps, balanced trade, all labor is the same) led to the need for new theories (e.g., New Trade Theory, heterogeneous firms).
- **Modern Relevance:** Ricardian models (like Eaton-Kortum) and extensions of HO remain vital for explaining **aggregate trade flows** and **distributional effects**.

Sources

- Dave Donaldson lecture notes
(<https://dave-donaldson.com/teaching/>)
- Isabelle Mjean lecture notes
- Treb Allen lecture notes (<https://sites.google.com/site/treballen/undergraduate-trade/>)
- Gianluca Benigno lecture notes (Background Note on Trade and Trade Imbalances)
- Ben Shepherd - The gravity model of international trade
- Costas Arkolakis - A Primer on Gains From Openness
- Saylor Academy: The Ricardian Theory of Comparative Advantage
(https://saylordotorg.github.io/text_international-trade-theory-and-policy/s05-the-ricardian-theory-of-compar.html)